

3 — Functional & Operational User Guide

What this document is: what the platform actually does from a user's seat — every major area, who can access it, and how the day-to-day operational tasks are performed.

Verified against the running codebase on 2026-09-07.

3.1 What this platform is

Three things that reinforce each other:

- 1. **The book** — *Transforming Healthcare: A Five-Pillar Framework with Vermont as the National Proving Ground*. The intellectual foundation.
- 2. **The platform** — pillar intelligence hubs, a Research Lab of interactive analytical tools, state dashboards, and an AI Analyst grounded in the book's evidence base.
- 3. **The Academy** — structured courses, tracks, and certifications.

The organising idea running through all three: **five pillars** (Policy → Technology → Economics → Clinical → Operations), each held to a cross-cutting **Equity Imperative** ("is it just?").

3.2 Getting in

The beta gate

The site currently sits behind a beta gate. A valid code (checked against the `beta_access_codes` table) sets a cookie and unlocks the site. Codes can be scoped to a specific domain.

Accounts and tiers

Sign up at `/signup`, log in at `/login`, reset via `/forgot-password`.

Tier	Price	What it adds
Free	—	Public content, limited AI
Student	\$19/mo · \$180/yr	Academy courses and certifications
Subscriber	\$29/mo · \$279/yr	All articles, reports, AI Analyst, dashboards
Professional	\$99/mo · \$959/yr	Certification tracks, CME credits
Team	\$23/seat/mo (2–25 seats)	Subscriber access, centrally billed
Advisory / Admin	Internal	Advisory tooling; admin console

Access is strictly hierarchical — each tier includes everything below it. Hitting a gated feature redirects to [/upgrade](#). Billing is managed at [/account](#) and [/billing-policy](#) (Stripe customer portal).

3.3 The five pillar hubs

Each pillar has an intelligence hub with topic cards, tools, courses, latest reports, and a "From the Book" callout linking to the relevant chapters.

Hub	Question it answers	Path
Policy	Is it permissible?	/policy
Technology	Is it possible?	/technology
Economics	Is it sustainable?	/economics
Clinical	Is it effective?	/clinical
Operations	Is it executable?	/operations
The Equity Imperative	Is it just?	/equity

The Equity Imperative is deliberately **not** presented as a sixth pillar — it appears as a cross-cutting test applied to all five, in the navigation, on the framework map, and in the scoring tools.

[/about/framework](#) is the interactive Five-Pillar Map: five nodes, nine directed dependencies, with the Equity Imperative drawn as the ring enclosing them. Click any pillar to trace what it enables, drives, or requires.

3.4 The Research Lab

[/research-lab](#) — 39 registered interactive tools, grouped into benches:

Bench	Contains
policy-quality	Policy Simulator, H.R. 1 Cliff, Work Requirements Calculator, Hospital Stress Test, Clinical Quality Optimizer
payment-models	APM Design Lab, Global Budget Modeler, Shared Savings Calculator, CEA Calculator
interoperability	FHIR Lab, Risk Stratification Engine, EMR/EHR Lab, Statewide EHR Modeler
technology-ai	AI Clinical Governance Lab, Digital Health Lab

Bench	Contains
vbc-clinical-quality	Clinical Data Exchange, VBC Quality Measures, High vs. Low Value Care
population-equity	Health Equity Studio (HEROI) , Population Health Modeler — the Equity Imperative's cross-cutting group
knowledge-workspace	Transformation Scorecard, VBC Readiness Assessment, Workforce Modeler, CIN & EMS Modelers, Evidence Library

Top-level simulators and dashboards

- [/htr-simulator](#) — score a transformation scenario across the five pillars, with the Equity Imperative applied as a justice check.
- [/impact-simulation](#) — model a scenario (hospital restructuring, global budgets, Hospital-at-Home, EMS regionalization) and see it propagate.
- [/transformation-friction-index](#) — quantify implementation barriers.
- [/hti-dashboard](#) — Health Transformation Index scoring engine.
- [/dashboard](#), [/states](#), [/compare-states](#) — 50-state performance data.
- [/bed-capacity](#) — live Vermont bed availability and transfer routing.
- [/investment-tracker](#) — M&A, VC, and PE activity.

3.5 The AI Analyst

Available as a **sidebar widget** throughout the site and as a **full chat** at [/chat](#).

What it's good at

- Vermont operational questions — it calls live data tools for hospital financials, bed capacity, Act 167 recommendations, transfer routing, and HSA population data.
- Policy and framework questions grounded in the book and platform content, with citations.
- Pointing you to the right tool or page for a task.

How it behaves

- Classifies your question first (Vermont-operational / Vermont-policy / national), then answers accordingly.
- Answer quality scales with your tier — higher tiers route to stronger models.
- It is instructed never to fabricate statistics or citations; if it doesn't know, it should say so and point to a resource.

Expected quirk: the very first message after a quiet period may take 5–15 seconds while the backend wakes from sleep. Subsequent messages are fast. This is a deliberate cost tradeoff.

3.6 The Academy

[/academy](#) — 14 courses, all with rich content.

Area	Path
Course catalog / tracks	/academy/tracks
Getting started (role-personalized)	/academy/getting-started
Personalized learning	/academy/personalized-learning
Case studies	/academy/case-studies
Webinars	/academy/webinars
Glossary	/academy/glossary
Faculty	/academy/faculty

Courses include: Medicaid 101 · Value-Based Care · Health Equity & SDOH · AI & Machine Learning in Healthcare · Interoperability & Data Exchange · Population Health Management · Medicare Fundamentals · Behavioral Health Integration · Revenue Cycle Management · Hospital Finance · Clinical Quality Measurement · Medicaid Managed Care Operations · Genomics & Precision Medicine · HIE & Health Reform (onboarding).

Learner features: progress tracking, quizzes with passing scores, bookmarks, lesson notes, audio slots, and certificates ([/api/academy/certificates](#)).

3.7 The book on the platform

Feature	Path
Book landing page + chapter browser	/book
Reader mode (chapter by chapter)	/read/[slug]
Audio narration	/book/listen
Saved chapters and notes	/saved , /library

Every chapter maps to platform tools through "Work This Chapter on the Platform" tables, and pillar pages carry "From the Book" callouts pointing back to the relevant chapters.

[frontend/lib/taxonomy/chapters.ts](#) is the single source of truth for that mapping.

3.8 Other user-facing areas

Area	Path	Notes
The Wire	/the-wire	News feed with comments and upvotes
Community	/community	Threads, posts, upvotes
Connect	/connect	Peer cohorts, expert office hours, pillar circles, grant finder
Advisory	/advisory	8 practice areas — hidden on the review brand
Vermont programs	/vermont-act-167 , /vermont-act-68 , /vermont-medicaid , /vermont-blueprint , /vermont-rht-program , and more	Program deep-dives
Medicaid eligibility simulator	/medicaid-eligibility-simulator	Step-through eligibility screening
Search	/search	Site-wide
Admin console	/admin	Beta codes, user management
System vitals	/system-vitals	Platform health
Sanity Studio	/studio	Content editing (embedded)

3.9 Operational runbook — routine tasks

Publish new editorial content

1. Open </studio>, create/edit the document, publish.
2. The Sanity webhook fires ingestion automatically, so the AI Analyst picks it up. Verify with `curl https://vhp-backend.fly.dev/api/ingest/status`.
3. If the AI still can't find it, that content type may not be in `SANITY_QUESTIONS` (`backend/services/indexing.py`).

Add or update an Academy lesson

1. Write the lesson body in Sanity (`academyModule`).
2. Ensure the Supabase lesson row's `sanity_slug` matches the Sanity slug — otherwise learners see thin placeholder content:

```
cd frontend
node scripts/link-sanity-slugs.mjs      # dry run
node scripts/link-sanity-slugs.mjs --commit
node scripts/audit-courses.mjs         # verify rich/total
```

Push course structure changes live

Course structure lives in **frontend/content/*.json**. After editing:

```
cd frontend
node scripts/seed-all-courses.mjs      # upsert-only, safe to re-run
```

This writes to the **production** database. It never deletes — removing a course or lesson means deleting Supabase rows directly.

Rebuild the AI's knowledge index

```
curl -X POST https://vhp-backend.fly.dev/api/ingest \
  -H "Authorization: Bearer $INGEST_SECRET"
```

Run after backend deploys, bulk content imports, or embedding model changes.

Grant or change a user's access

Roles live in the **user_roles** table; a user gets the **highest** role they have a row for. Paid roles are normally set automatically by the Stripe webhook. Manual overrides go through **/admin** or a direct DB update.

Issue beta access codes

/admin/beta-codes (or the **beta_access_codes** table). Codes can be domain-scoped.

Publish a new edition of the book

1. **./book.sh check** — surface any Google Docs edits not yet in the Markdown.
2. Fold those edits into **HTR_Book_v42.md**.
3. **python3 book-build/check_manuscript.py** — must report 0 errors.
4. **./book.sh build** — regenerate the **.docx**.
5. Upload that **.docx** to Google Docs → Download as PDF → save over **HTR_Book_v42.pdf**.
6. Copy the PDF to **frontend/public/** so **/book** and **/read** serve the current edition.

Verify the PDF is actually current before wiring it in — a stale export has shipped before.

3.10 Deploying changes

Change to	How it ships
Anything under frontend/	Push to main → Vercel auto-deploys
Anything under backend/	Push to main → GitHub Action deploys to Fly
Database schema	Manual — apply the migration in Supabase

Change to	How it ships
Content (Sanity)	Publish in Studio — live immediately
Course structure	Run the seed script manually

Before pushing, run the local gate:

```
cd frontend && npm run smoke # typecheck + lint + build + bundle budget
```

3.11 If something looks wrong

What you see	Likely cause
First AI message is slow	Backend cold start — normal
AI gives vague answers, misses recent content	Index not rebuilt, or OpenAI credit exhausted
Academy lesson looks thin/placeholder	<code>sanity_slug</code> not linked
Paid feature still locked after payment	Stripe webhook didn't land — check <code>stripe_events</code>
Advisory section missing	Expected on the <code>review</code> brand domains
New Sanity content invisible to the AI	Content type not in the ingestion queries

Quick health checks

```
curl https://vhp-backend.fly.dev/health
flyctl status -a vhp-backend
cd frontend && node scripts/audit-courses.mjs
```